8.6 Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad \text{(XR8.6a)}$$

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

a. We are curious whether holding education, experience, and *METRO* constant, there is the same amount of random variation in wages for males and females. Suppose $\text{var}(e_i|\mathbf{x}_i, FEMALE = 0) = \sigma^2_M$ and $\text{var}(e_i|\mathbf{x}_i, FEMALE = 1) = \sigma^2_F$. We specifically wish to test the null hypothesis $\sigma^2_M = \sigma^2_F$ against $\sigma^2_M \neq \sigma^2_F$. Using 577 observations on males, we obtain the sum of squared OLS residuals, $SSE_M = 97161.9174$. The regression using data on females yields $\hat{\sigma}_F = 12.024$. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

b. We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose $\text{var}(e_i|\mathbf{x}_i, MARRIED = 0) = \sigma^2_{SINGLE}$ and $\text{var}(e_i|\mathbf{x}_i, MARRIED = 1) = \sigma^2_{MARRIED}$. Specify the null hypothesis $\sigma^2_{SINGLE} = \sigma^2_{MARRIED}$ versus the alternative hypothesis $\sigma^2_{MARRIED} > \sigma^2_{SINGLE}$. We add *FEMALE* to the wage equation as an explanatory variable, so that

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad \text{(XR8.6b)}$$

Using $N = 400$ observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

c. Following the regression in part (b), we carry out the NR^2 test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

e. The OLS fitted model from part (b), with usual and robust standard errors, is

$\widehat{\text{WAGE}} =$	-17.77	$+\, 2.50EDUC$	$+\, 0.23EXPER$	$+\, 3.23METRO$	$-\, 4.20FEMALE$
(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

f. If we add *MARRIED* to the model in part (b), we find that its t-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

(a)

$$df_M = 577 - 3 - 1 = 573$$

$$df_F = 1000 - 577 - 3 - 1 = 419$$

$$\hat{\sigma}^2_M = SSE_M/df_M = 97161.9174/573 = 169.567$$

$$F = \hat{\sigma}^2_M/\hat{\sigma}^2_F = 169.567/12.024^2 = 1.173$$

$$RR:\{F > F_{(0.025,573,419)}\ or\ F < F_{(0.925,573,419)}\}$$

由於樣本數極大，臨界值 $F_{(0.025,573,419)}$ 的精確值約為 1.18（可透過常態近似或查表得之）。

因為檢定統計量 1.173 小於上界臨界值，我們無法拒絕虛無假說。結論是，在 5% 顯著水準下，沒有足夠的證據表明男女的工資變異數存在顯著差異。

(b)

$$K = 4 + 1 = 5$$

$$MSE_{SINGLE} = SSE_{SINGLE}/df_{SINGLE} = 56231.0382/(400 - 4 - 1) = 142.357$$

$$MSE_{MARRIED} = SSE_{MARRIED}/df_{MARRIED} = 100{,}703.0471/(1000 - 400 - 4 - 1) = 169.249$$

$$F = MSE_{MARRIED}/MSE_{SINGLE} = 169.249/142.357 = 1.189$$

$$RR:\{F > F_{(0.05,595,395)} \approx 1.15\}$$

因為 1.189 > 1.15，我們拒絕虛無假說。結論是，有統計證據支持已婚人士的工資變異確實顯著大於單身人士。

(c)

檢定結論：輔助迴歸中包含了 4 個解釋變數（$EDUC, EXPER, METRO, FEMALE$），因此此卡方檢定統計量的自由度為 4。

在 5% 顯著水準下的臨界值為 $\chi^2(0.05,4) = 9.488$。因為 59.03 遠大於 9.488，我們強烈拒絕同質變異性（Homoskedasticity）的虛無假說，確認模型中存在異質變異性。

對 (b) 部分的證據效力：沒有提供直接證據。因為在這個 NR^2 檢定的輔助迴歸方程式中，並沒有包含 $MARRIED$ 這個變數。該檢定只能證明殘差變異數與（$EDUC, EXPER, METRO, FEMALE$） 有關，無法說明殘差變異數是否與婚姻狀況$MARRIED$ 相關。

Based on the NR^2 test statistic of 59.03, we can conclude that there is evidence of heteroskedasticity in the regression model at the 5% significance level. The NR^2 test evaluates whether the variance of errors between observations in the regression model is constant, but it does not specifically examine whether the error variation differs between married and unmarried individuals. Therefore, although the result of the NR^2 test suggests potential heteroskedasticity, it does not directly provide evidence regarding the specific issue discussed in the question.

(d)

df = number of variables + number of variables square + number of interactions - indicator variables = 4 + 4 + 6 - 2 = 12

Rejection region

$$\{\chi: \chi^* > \chi_{(12,0.95)} = 21.02607\}$$

test statistic

$$\chi^* = 78.82$$

$$\chi^* = 78.82 > \chi_{(12,0.95)} = 21.02607$$

⇨ Reject H0

We have significant evidence to reject homoskedasticity. It means

$$\sigma^2_{SINGLE} \neq \sigma^2_{MARRIED}$$

(e)

變窄的估計（穩健標準誤 < 一般標準誤）：

$$EXPER\ (0.029\ <\ 0.031)$$

$$METRO\ (0.84\ <\ 1.05)$$

$$FEMALE\ (0.80\ <\ 0.81)$$

變寬的估計（穩健標準誤 > 一般標準誤）：

$$截距項\ (2.50\ >\ 2.36)$$

$$EDUC\ (0.16\ >\ 0.14)$$

當模型存在異質變異性時，一般的 OLS 標準誤會產生偏誤，但這個偏誤的方向（高估或低估）並非固定的，而是取決於殘差變異數與解釋變數平方和交叉項的相關性結構。因此，穩健標準誤比一般標準誤大或小都是正常的數學現象，並不構成矛盾。

(f)

這兩者是完全相容的。此處的 t 檢定 ($t\ \approx\ 1$) 測試的是 $MARRIED$ 對工資的條件期望值（平均數）是否有顯著影響。t 值為 1 代表不顯著，也就是婚姻狀況並未使工資水準產生系統性的平移。然而，(b) 部分探討的是 $MARRIED$ 對工資的變異數（資料的離散程度）是否有影響。一個變數完全可以「不改變平均水準的同時，擴大或縮小資料的波動範圍」。例如，已婚人士平均賺得不比單身人士多，但由於職業選擇更多元，有人賺極多、有人賺較少，導致整體變異變大。因此，平均數檢定不顯著與變異數檢定顯著兩者並不互斥。

8.16 A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

MILES is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

a. Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.

b. Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?

c. Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.

d. Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?

e. Obtain GLS estimates assuming $\sigma_i^2 = \sigma^2 INCOME_i^2$. Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

(a)

$$MILES = \beta_1 + \beta_2 \, INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

$$MILES = -391.548 + 14.201 \; INCOME + 15.741 \, AGE + -81.826 \, KIDS + e$$

95% interval estimate for the effect of one more child on miles traveled

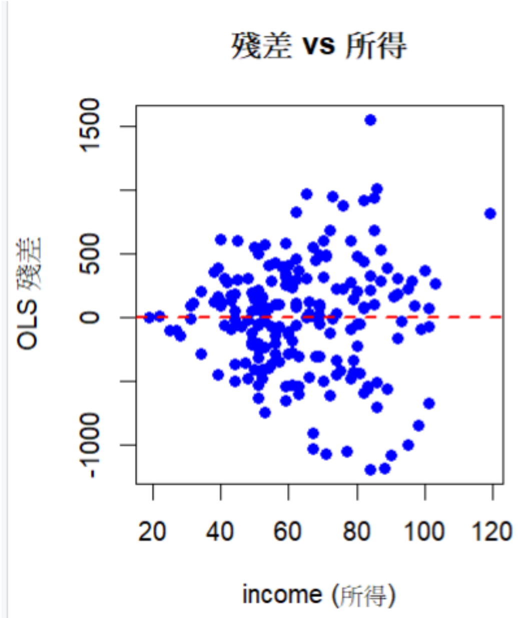
$$[\, -135.3298 \, , -28.32302 \,]$$

```
Call:
lm(formula = miles ~ income + age + kids, data = vacation)

Residuals:
     Min       1Q   Median       3Q      Max 
-1198.14  -295.31    17.98   287.54  1549.41 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -391.548    169.775  -2.306   0.0221 *  
income        14.201      1.800   7.889 2.10e-13 ***
age           15.741      3.757   4.189 4.23e-05 ***
kids         -81.826     27.130  -3.016   0.0029 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 452.3 on 196 degrees of freedom
Multiple R-squared:  0.3406,	Adjusted R-squared:  0.3305 
F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
> #  建構 KIDS 變數的 95% 信賴區間
> ci_kids <- confint(model_a, "kids", level = 0.95)
> 
> ci_lower <- ci_kids[1]
> ci_upper <- ci_kids[2]
> 
> #  最終結果
> cat("95% CI: [", ci_lower, ",", ci_upper, "]\n")
95% CI: [ -135.3298 , -28.32302 ]
```

(b)

觀察到明顯的異質變異，變異主要與 income 有關



1. 觀察「殘差 vs 所得 (income)」

資料點呈現一個明顯向右開口的漏斗狀(funnel shape)。

當所得較低（例如 20-40 區間）時，殘差緊密地聚集在紅色的水平虛線 (Y=0) 附近，代表誤差的變異數很小；但隨著所得增加（例如 80-100 區間），殘差的上下分布範圍變得非常寬廣，代表誤差的變異數顯著變大。

結論：這在現實中很合理。低所得家庭的旅遊預算有限，他們的旅行距離通常都被限制在較小的範圍內；而高所得家庭的選擇彈性大很多，他們可能去很遠的地方（產生極大的正殘差），也可能選擇在家附近的高級飯店度假（產生負殘差），因此行為變異度大很多。

2. 觀察「殘差 vs 年齡 (age)」

殘差點在整張圖表上呈現隨機散佈 (random scatter)。

無論家庭成年成員的平均年齡是 25 歲還是 55 歲，殘差上下的波動幅度（寬度）看起來都差不多。這表示誤差項的變異數並沒有隨著年齡的改變而有系統性的放大或縮小。

結論：在這個變數上，我們沒有觀察到違反同質變異假設的證據。

(c)

$$H_0:\ \sigma_1^2 \ = \ \sigma_2^2$$

$$H_0:\ \sigma_1^2 \ < \ \sigma_2^2$$

將總共 200 筆觀察值依照 INCOME 遞增排序。

低所得組 (Group 1)： 前 90 筆資料 ($N_1 = 90$)

高所得組 (Group 2)： 最後 90 筆資料 ($N_2 = 90$)

$$K = 4$$

$$df_1 \ = \ N_1 - K \ = \ 90 \ - \ 4 \ = \ 86$$

$$df_2 \ = \ N_2 - K \ = \ 90 \ - \ 4 \ = \ 86$$

test statistic

$$GQ = \frac{SSE_2 / \ (N_2 - K)}{SSE_1 / \ (N_1 - K)}$$

Critical Value $RR: F\{0.05, 86, 86\} \approx \ 1.43$

$GQ = 3.104061$

$GQ \in RR$

⇨ Reject H0

在 5% 的顯著水準下，有充分的統計證據表明該模型存在異質變異 (Heteroskedasticity)，且誤差變異數確實隨著家庭所得的增加而擴大。

```
> cat("--- (c) Goldfeld-Quandt 檢定 ---\n")
--- (c) Goldfeld-Quandt 檢定 ---
> cat("GQ 統計量:", gq_stat, "\n")
GQ 統計量: 3.104061
> cat("P-value:", p_val_gq, "\n\n")
P-value: 1.64001e-07
```

(d)

95% interval estimate for the effect of one more child on miles traveled

$$[-139.323, -24.3298]$$

將 (d) 的穩健信賴區間與 (a) 的傳統 OLS 信賴區間相比，我們可以觀察到 (d) 的信賴區間變得更寬了（The robust interval is wider）。

同時，KIDS 變數的穩健標準誤大於傳統 OLS 的標準誤。

```
> cat("95% CI: [", ci_lower_d, ",", ci_upper_d, "]\n\n")
95% CI: [ -139.323 , -24.32986 ]
```

(e)

比較 1：GLS vs. (a) 傳統 OLS

差異： 不僅信賴區間的寬度不同，連 $\widehat{\beta_4}$ 的點估計值 (Point Estimate) 也改變了。

原因： OLS 對所有資料一視同仁；但 GLS 給予低所得家庭（變異小、資訊精確）較大的權重，給予高所得家庭（變異大、雜訊多）較小的權重。因此 GLS 算出來的係數會與 OLS 不同，且理論上 GLS 具備更高的效率 (Efficiency)，是最佳線性不偏估計量 (BLUE)。

比較 2：傳統 GLS vs. (d) 穩健 OLS

差異：(e) 的傳統 GLS 信賴區間，會比 (d) 的穩健 OLS 信賴區間更窄 (Narrower)。

原因： (d) 的穩健 OLS 只是消極地承認異質變異並放大標準誤來涵蓋風險；而 (e) 的 GLS 則是積極地利用已知的變異數結構（與 $INCOME^2$相關）去修正模型。因為 GLS 使用了更多資訊，所以它估計得更精準，標準誤更小，信賴區間更窄。

比較 3：傳統 GLS vs. 穩健 GLS

差異： 觀察這兩者的區間差異是否很大。

原因： 如果傳統 GLS 與穩健 GLS 的區間非常接近，代表我們一開始假設的變

異數形式 $Var(e_i) = \sigma^2\ INCOME_i^2$ 是非常準確的；如果兩者差異還是有點大，代表雖然我們做了 GLS 轉換，但殘留的誤差可能還有一點點其他的異質變異結構沒有被捕捉到，此時看「穩健 GLS」會是最保險的結論。

```
--- (e) GLS 結果 ---
> cat("傳統 GLS 95% CI: [", ci_lower_e_trad, ",", ci_upper_e_trad, "]\n")
傳統 GLS 95% CI: [ -119.8945 , -33.71808 ]
> cat("穩健 GLS 95% CI: [", ci_lower_e_robust, ",", ci_upper_e_robust, "]\n")
穩健 GLS 95% CI: [ -121.4134 , -32.19919 ]
```